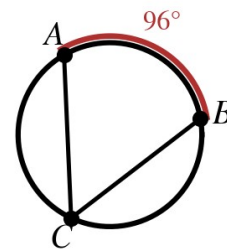


SC Honors Geometry - Nappier
Homework 10.4 – Inscribed Angles and Polygons

Name: _____

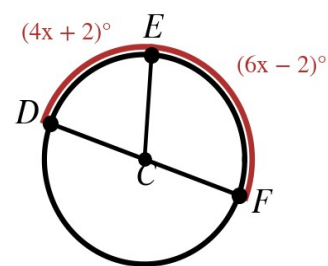
Inscribed angles and arcs.

1. An inscribed angle intercepts a 120° arc. Find the angle.
2. An inscribed angle measures 36° . Find its intercepted arc.
3. In the TOP figure, $m\widehat{AB} = 96^\circ$. Find $m\angle ACB$.
4. Two inscribed angles intercept the same arc. One measures 39° . Find the other.



Solve for the variable.

5. A triangle is inscribed in a semicircle; its acute angles are $(4x + 6)^\circ$ and $(5x + 3)^\circ$. Find x and both angles.
6. An inscribed quadrilateral has opposite angles $(3x + 4)^\circ$ and $(2x + 6)^\circ$, and the other pair $(4y - 15)^\circ$ and $(y + 30)^\circ$. Find x and y .
7. In the BOTTOM figure, DF is a diameter and the two arcs above it measure $(4x + 2)^\circ$ and $(6x - 2)^\circ$. Find x and each arc.



Think it through.

8. An inscribed angle intercepts a semicircle. What is its measure and why?
9. A regular pentagon is inscribed in a circle. Find the arc between neighboring vertices.
10. An inscribed angle measures 54° . Find the central angle that opens onto the same arc.